

Strategic Business Objectives

- **Reducing Claims Settlement Time:** Reduce the total time from the occurrence of an insurance claim to the closure of the case and identify specific operational delays at key stages of the process.
- **Evaluate RPA effectiveness and optimize RPA:** Assess the proportion of automated cases and compare the speed and consistency of processing between the robotic system and human agents at various stages.
- **Improving underwriting and risk calculation accuracy:** Identify the most loss-prone segments of the portfolio based on an analysis of how payout amounts correlate with customer age, accident types, policy types, and vehicle make, model, and year of manufacture.
- **Optimizing the operational workload of adjusters:** Analyze the productivity of top adjusters, compare their workloads, approved claim amounts, and average decision-making speed.
- **Monitoring key high-level metrics:** Provide C-level management with a single dashboard to monitor the total volume of claims payments, the number of cases, the average severity of losses, and the trend in claims payments over time.

Key Analytical and Technical Tasks

Operational Analysis and Process Mining

- Calculate the total duration of the settlement cycle and the duration of individual stages in hours and days.
- Calculate the time delays between key process checkpoints.
- Compare the average and median processing times for the RPA and Human categories, and assess the variability (distribution) of claim settlement times.

Financial, Product, and Risk Analysis

- Calculate the total and average claim payouts by policy type and car make.
- Build a two-dimensional loss matrix by accident type and policy type.
- Create a breakdown of the most loss-prone car makes and models, broken down by year of manufacture.
- Examine the distribution of the number of claims and the average claim amount by customer age group.
- Evaluate the structure and trends of accident types by year in a 100% cumulative chart.

Staff Performance Analysis

- Create a performance matrix for the top 10 claims adjusters.
- Calculate the following for each claims adjuster: the number of claims processed, the total amount of payouts, and the average decision-making time.

Conclusions

Operational Delays and Process Mining

Key Bottleneck: The initial notification of a claim to the assignment of an adjuster (FNOL - Assign) takes an average of 15.01 days. This is over 42% of the entire settlement cycle (Avg Cycle Time = 35.01 days).

Decision Speed: Once a case is assigned to an adjuster, the Assign - Decision process is significantly faster - an average of 5.01 days.

Robotic Automation Effectiveness (RPA vs Human)

RPA Penetration Rate: The RPA Adoption Rate reached 35.33%, indicating a successful initial implementation of the technology.

Comparative Speed: The overall median case closure time (Cycle Time) for a robotic system and a human is almost the same (about 35 days).

Variability and stability: Time distribution shows that RPA has significantly fewer anomalous deviations compared to manual processing. The robot works more predictably, but does not yet speed up the overall cycle.

Adjuster Performance

Load and uniformity: The top 10 adjusters process from 9 to 15 cases during the study period.

Speed of decision: The average decision time (Avg Time to Decision) fluctuates in a narrow range - from 19 to 22 days, which indicates a balanced distribution of complex cases among the assessors.

Risk profile and underwriting (Risk & Underwriting)

The most unprofitable segments: The largest volume of payments falls on mass segment car models: Ford F-150, Toyota Camry, Honda Civic and Chevrolet Silverado.

Age distribution: The number of insured events is consistently high among drivers aged 20 to 80, however, the highest average payout amounts are observed in the age groups 50 - 55 and 75 - 80.

Practical Business Recommendations

Process Optimization and Cycle Time Reduction

Implement Auto-Assign: Eliminate the main FNOL - Assign bottleneck, which currently takes 15 days, by automating case assignment using routing algorithms. This will reduce the total resolution time by 30 - 40% (to 20 - 22 days).

Reduce Reserve Setup Time: Set a strict SLA (within 24 hours) for establishing the insurance reserve after an appraiser is assigned to improve liquidity management.

Scaling and Refining RPA

Revise RPA logic during decision-making stages: Since the median RPA processing time is equivalent to that of a human, the robot's algorithms must be reviewed. Switch to 100% automated approval for minor claims (e.g., up to \$2,000) without requiring a wait for system delays.

Increase the RPA Adoption Rate to 50%+: Fully automate the "Set Reserve" and "Payment Sent" stages for standard policies (Collision, Comprehensive).

Underwriting and Rate Schedule Review

Adjustment of Coefficients for Top Models: Increase base insurance rates or deductibles for car models with the highest total loss ratios (Ford F-150, Honda Civic, Toyota Camry, model years 2018 - 2022).

Age-based risk segmentation: Take into account the increased average claim amount in the 50 - 55 and 75+ age groups when calculating individual client scoring.